

Model review and revisions

JMP progress, September 17–October 1, 2026

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I reviewed the full model and calibration: preferences, borrowing rules, demographic accounting, equilibrium conditions, empirical measurements, targets, and the estimation procedure.

- Returned to a Stone–Geary housing requirement that applies only to parents, retaining nonlinear benefits from children.
- Reviewed equivalence scales and housing-share alternatives.
- Rebuilt the first-birth housing evidence and national input measures.
- Checked demographic accounting, credit, and adult entry; recalibrated the model.

Household decisions

Overlapping generations of households differ in age, earnings, wealth, housing tenure, and family size.

- During fertile ages, households choose whether to attempt a birth; success becomes less likely with age.
- They choose consumption, saving, housing size, and whether to rent or own; renters pay rent, while owners make a down payment and face selling costs.
- Persistent earnings risk and borrowing limits shape saving; a first-birth fixed cost and taste shocks shape fertility.
- Households retire, face mortality risk, and value bequests.

The interaction of housing costs and available wealth affects both birth timing and tenure choice.

Children and housing needs

The first child requires basic space at home; each additional child raises the goods needs of a larger family.

$$u(c, h, m, t) = \frac{\left[\frac{c^{\alpha_0} s_t(h, m)^{1-\alpha_0}}{e(m)} \right]^{1-\sigma}}{1-\sigma} + \psi m^{1-\gamma},$$
$$e(m) = \left(\frac{2 + 0.7m}{2} \right)^{0.7}, \quad s_t(h, m) = \eta_t [h - h_P \mathbf{1}\{m > 0\}],$$
$$\eta_{\text{rent}} = 1, \quad \eta_{\text{own}} = \chi.$$

Here h is physical rooms, m is children at home, and t indexes renter or owner tenure. Parents choose $h > h_P$, and α_0 is the constant consumption exponent. The equivalence scale raises goods needs with family size; the benefit is zero at $m = 0$, with ψ setting its level and γ its curvature so the marginal benefit diminishes as m grows.

Demographic and market equilibrium

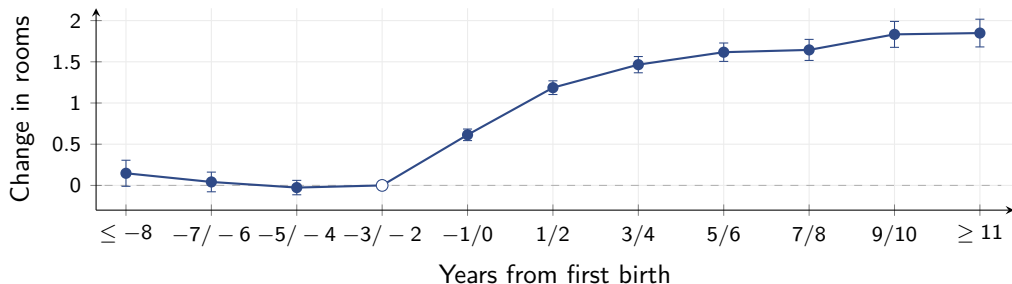
Children leaving home and new adult households entering the economy are separate transitions.

- Children leave dependency independently; adult entry follows earlier births with a lag.
- Positive estates finance entrant assets; any remainder leaves the economy.
- The house price adjusts to satisfy birth renewal; population scale clears aggregate housing demand against supply.
- A balanced payroll tax finances equal retiree pensions.

A stationary equilibrium has constant prices, allocations, and cohort structure, with demographic reproduction and market clearing.

Housing around the first birth

I rebuilt the PSID event study using interview dates and household status before the first birth.



At years 3–4, rooms rise by **1.465** (SE 0.050), ownership by **26.886 pp**, and recent moving falls by **21.147 pp**.

Reference person or spouse before birth. Baseline: years $-3/-2$; person and year effects; survey weights; 95% pointwise intervals.

Calibration strategy and inputs

I estimate a 2007 stationary economy using national measures of housing, earnings, wealth, and fertility.

Input	Value	Measurement
Mean occupied rooms	5.729	AHS 2007
Pension / working-household earnings	0.229	CPS ASEC, income 2007
Annual child-directed estates / wealth	0.729%	SCF 2007 mortality proxy
Annual depreciation / property tax	1.416% / 1.060%	National housing inputs

- Ten parameters are estimated jointly, including ψ ; housing supply scale H_0 is fixed. Other fixed inputs include earnings and demographics, housing depreciation and taxes, the interest rate, and mortgage financing.
- The objective is a weighted sum of squared gaps for ten moments, using working error scales. I increased the age-25 fertility weight fourfold in this search.
- Three further moments are validation checks. Completed fertility is reported separately as a renewal check.

For each parameter vector, I solve household choices, birth renewal, and housing-market equilibrium. The first-birth housing comparison uses a model proxy for the empirical event study.

Stone–Geary calibration

2007 stationary economy; current estimate

Moment	Target	Model
Completed fertility (renewal check)	2.100	2.100
Childlessness, ages 40–44 (%)	19.828	20.253
Exactly one child among mothers, ages 40–44 (%)	21.366	21.063
Mean first-birth age (years)	25.976	25.956
First births at age 30+ (%) [†]	24.928	22.931
Wealth / earnings	6.927	6.489
Annual child-directed bequest flow / wealth (%)	0.729	0.666
Older wealth/income p90/p50 [†]	3.516	2.976
Mean occupied rooms	5.729	5.977
Ownership, ages 30–55 (%)	67.626	65.697
First-birth rooms difference	1.465	1.223
Family rooms difference [†]	0.385	0.300
Recent-parent ownership difference (pp)	12.761	11.837
Children per woman at age 25, capped at three	0.810	0.531

[†]Validation moments. Ten scored moments; completed fertility is a separate renewal check. Numerical repeat verified; search convergence not established.

Calibrated parameters

Ten parameters are estimated jointly, including ψ .

Parameter	Economic interpretation	Value
β	Annual discount factor	0.967
χ	Owner housing-service premium	1.093
F_1	First-birth fixed utility cost	0.401
κ_1	First-birth taste dispersion	0.131
κ_+	Later-birth taste dispersion	0.364
θ_0	Bequest weight	0.079
γ	Child-benefit curvature	0.097
κ_T	Tenure-choice dispersion	0.013
ψ	Child-benefit level	0.172
$h_p^\dagger$	Basic space requirement for parents (rooms)	2.300

Fixed inputs: $H_0 = 6.294$; $\sigma = 2$; $\alpha_0 = 0.733$; annual interest 2%; mortgage-financed share 80%. $^\dagger h_p$ is at its upper bound. Birth-taste dispersions and bequest weight are near their lower bounds; full bounds are in the saved tables.

Preference alternatives

I compared different ways of making children raise household needs, especially housing needs at the first birth.

- The Stone–Geary requirement captures basic space for parents; the equivalence scale captures the cost of a larger family.
- I tested child-dependent housing shares, with and without a compensating utility adjustment, to raise the housing budget share at the first birth.
- I softened the housing requirement to check the role of the sharp threshold. I also considered keeping only the equivalence scale, with constant housing preferences.
- The nonlinear child benefit allows the marginal benefit to diminish as the number of children at home rises.

Room responses alone are insufficient; I assess alternatives jointly with fertility timing, wealth, and tenure.

Credit, initial wealth, and remaining fit

- I reviewed mortgage purchase and sale decisions and unsecured borrowing limits, distinguishing the down payment needed to own from the liquidity needed to have a child.
- I compared zero, empirical, and nonnegative entrant wealth rules at a common 2% annual real interest rate.
- The current Stone–Geary estimate uses nonnegative entrant wealth with the original mean and allows no unsecured borrowing.
- The main remaining fit problem is fertility at age 25: 0.531 children ever born per woman, against a target of 0.810 (both capped at three).
- The estimated parent space requirement, $h_P = 2.3$, is at its upper bound, so I still need to check whether the search bound is restricting the fit.